

# Joana Mizrahi

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Robotics, Controls & Autonomy | Stanford M.S. Electrical Engineering, graduating June 2027

## EDUCATION

**Stanford University**, Stanford, California June 2027

*M.S. in Electrical Engineering, depth in Control and Optimization* GPA 4.10

- Coursework: Experimental Robotics; Optimal & Learning-Based Control; Robot Perception; State Estimation & Filtering; Convex Optimization; Computational Imaging; Decision Making under Uncertainty; Deep Learning
- Projects: violin-playing robot on a Flexiv Rizon arm (CS 225A); vision-based vehicle tracking on KITTI with a particle filter and multi-cue likelihoods (AA 273); grasp-success prediction from 3D point clouds with PointNet (CS 230)

**Carnegie Mellon University**, Pittsburgh, Pennsylvania May 2025

*Exchange Program, Electrical and Computer Engineering* GPA 4.0

- One of 4 students selected for a 1-year exchange; grant from EPFL and Severin B. Hacker (Duolingo co-founder and CTO)
- Projects: embedded RTOS on STM32 (interrupts, scheduling, motor control); MEMS dual-axis micromirror; neural amplifier

**École Polytechnique Fédérale de Lausanne (EPFL)**, Lausanne, Switzerland May 2025

*B.Sc. in Microengineering* GPA 5.74/6

- Valedictorian: awarded the EPFL “Wafer d’Or” for the highest GPA in the Microengineering Bachelor’s program
- SHS Laureate: best Global Issues poster, 2023
- Projects: C++ robot simulation; embedded systems in AVR assembly; flexure-based marine chronometer oscillator

## EXPERIENCE

**Stanford Robotics Center**, Stanford, California June 2026 – September 2026

*Robotics Intern*

- Developed WorldNav, a framework for training outdoor navigation policies with on-policy RL inside a generative world model built from a single walkthrough video, for the Unitree Go2W quadruped
- Evaluation: extended the world model to generate terrain semantics together with each image, giving the policy a traversability reward that stays consistent with what it sees
- Exploration: bounded where the policy may train using measures of how well each generated view is supported by the recording, so it cannot exploit unreliable generations

**Stanford Robotics Center**, Stanford, California April 2026 – present

*Graduate Researcher. Mentor: Jing Liang. Research units with Prof. Jiajun Wu (CS 399) and Prof. Marco Pavone (AA 290)*

- Co-first author of the WorldNav paper (in preparation for CVPR 2027); built the initial pipeline that turns monocular video into a semantic 3D simulator, now evaluating the world model and policies on campus scenes
- Voice navigation on the Unitree Go2W: designed and implemented a real-time behavior-tree task planner with an OpenAI Realtime API voice interface and open-vocabulary hazard alerting (ROS 2)

**Virtual Traffic Lights**, Pittsburgh, Pennsylvania August 2024 – January 2026

*Software Engineer*

- Integrated traffic signal controller hardware (Econolite ASC/3, NTCIP over SNMP) with the company’s ML signal-timing optimizer for real-time closed-loop testing
- Built a backend service to version, schedule and approve AI-generated timing plans

## TEACHING

**Course Assistant**, AA228/CS238 Decision Making under Uncertainty, Stanford September 2026 – present

**Grader**, EE 102A Signals and Systems I, Stanford January 2026 – March 2026

**Teaching Assistant**, PHYS-101 Physics and MSE-101 Materials Science, EPFL September 2023 – June 2024

## SKILLS

**Programming:** Python, C/C++, MATLAB

**Robotics & ML:** ROS 2, PyTorch, reinforcement learning, video diffusion fine-tuning, Gaussian splatting, state estimation

**Software & hardware:** Git, Docker, gRPC; CATIA, Fusion 360, COMSOL, LTspice; LaTeX

**Languages:** French (native), English (fluent), German (basic)

## ACTIVITIES

Competitive badminton (national tournaments, CMU team); mathematics and French tutor (ARA)